

Puppet Stayman or not?

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Abstract

The Stayman convention offers an efficient tool for locating 4-4 major fits following a 1NT opening, enhancing the responder's ability to explore optimal contracts. This paper delves into the standard Stayman responses, rebids, and strategic considerations, while also introducing Puppet Stayman—a more nuanced variant designed to find 5-3 and 5-4 major-suit fits. Various hand configurations and example deals are used to evaluate the effectiveness of Stayman in different game contexts, and a large sample of deals are analyzed using double dummy solvers to give accurate results. While Puppet Stayman theoretically yields slight improvements in performance, its complexity and risks may outweigh the benefits for less experienced players or practical applications.

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1 Introduction

Stayman is a widely used bidding convention in bridge, typically employed by the responder after a 1NT opening to explore the possibility of a 4-4 fit in a major suit. The Stayman convention can be applied in various situations depending on the responder's hand distribution and high-card points (HCP). We will examine the variation of Stayman called Puppet Stayman, focusing on the its application and effectiveness.

2 Stayman

Stayman in the bridge convention commonly used in responder position against a natural no-trump opening.

West	North	East	South
2NT	/	3♣	/
?			

Here, 3♣ is a standard stayman asking bid (some systems promise a 4-card major, while some systems do not), which asks if partner has a 4⁺-card major (to find a 4-4 strain). The responses are 3♦ (none), 3♥ (4⁺-card in ♥), and 3♠ (4⁺-card in ♠ but not ♥). No other response is made.

Stayman is used in two cases: first, when the responder tries to find a major strain (either the normal 4-4 strain or a 5-3 strain by holding 5-4 majors, rebidding the longer suit at the second level after a potential 3♦), and second, when the responder expects to pass any response (garbage stayman). Some systems - ones that do not promise a 4-card major - use stayman as a transitional bid for 10⁺ HCP hands with 5⁺-card minors. (4-4 minors call 3♠. It is also common for only 5-5 minors to call 3♠ to find 5-4 strains for slam.) Other systems use minor transfers for length in one minor.

We will consider the simplest case: stayman promises a 4⁺-card major and 4⁺ HCP against a 20-21 HCP balanced 2NT. (2NT here does not promise stopper in 3⁺ suits.) For our analysis we will assume up to 27 HCP combined, for otherwise we are in possible slam (or slam interest) territory.

The logic overall here for stayman is simple: If we have only one major suit, we stayman and then raise to game. The same applies to 4-4 majors. If we have 5-4 majors, then we can respond after 3♦ and partner will either respond with 3NT or 4M. Additionally, we won't stayman with 4-3-3-3, since there is no trumping potential.

Deal 1 Example of rebidding after 3♦

♠ K Q 6		♠ J T 4 2
♥ K Q 6		♥ J T 5 3 2
♦ A K 7 4	W E	♦ Q 2
♣ K 9 6		♣ Q 2

West	North	East	South
2NT	/	3♣	/
3♦	/	3♥	/
4♥	/	/	//

Deal 2 Example of an implied 4-card major

♠ K Q 6 2		♠ J T 4 2
♥ K Q 6 2		♥ J T 5
♦ A K 6	W E	♦ Q 5 4 3 2
♣ K 3		♣ 2

West	North	East	South
2NT	/	3♣	/
3♥	/	3NT	/
4♠	/	/	//

Deal 3 We don't stayman with 4-3-3-3

♠ A K 9 6		♠ Q J 3 2
♥ K 9 6		♥ J 4 3
♦ A Q 6	W E	♦ J 3 2
♣ A 5 3		♣ Q 4 2

West	North	East	South
2NT	/	3NT	/
/	//		

3 Puppet Stayman

Puppet stayman is a little more complex, and are usually preferred in 2NT auctions, since many players don't open 1NT with good majors.

When partner calls stayman to your 2NT opening, reply with 3♦ (no 5+-card major, but has 4-card major), 3♥ (5+ ♥), 3♠ (5+ ♠ but not ♥), or 3NT (no 4+-card major).

The followup for 3♦ with a 4-card major is to call in the other major suit (to make partner declarer). With 4-4 majors or above, raises to, say, 4♣ or 4♦ may be used depending on the agreement (for different strengths).

Deal 4 Example of rebidding after 3♦

♠ K Q 9	W E	♠ J T 4 2
♥ K Q 9 2		♥ J T 5
♦ A K 7 4		♦ Q 3 2
♣ K 9		♣ Q 3 2

West	North	East	South
2NT	/	3♣	/
3♦	/	3♥	/
3NT	/	/	//

Deal 5 Example of normal response

♠ K Q 9	W E	♠ J T 4
♥ K Q 9 3 2		♥ J T 5
♦ A K 7		♦ Q 4 3 2
♣ K 9		♣ Q 3 2

West	North	East	South
2NT	/	3♣	/
3♥	/	4♥	/
/	//		

Deal 6 We bid 3♣ with 4-3-3-3 only to find a 5-4 strain.

♠ K Q 9	W E	♠ J T 4
♥ K Q 9 6 2		♥ J T 5 4
♦ A K 7		♦ Q 3 2
♣ K 9		♣ Q 3 2

West	North	East	South
2NT	/	3♣	/
3♥	/	4♥	/
/	//		

Deal 7 ... Continued

♠ K Q 9 6	W E	♠ J T 4
♥ K Q 9 6		♥ J T 5 4
♦ A K 7		♦ Q 3 2
♣ K 9		♣ Q 3 2

West	North	East	South
2NT	/	3♣	/
3♦	/	3NT	/
/	//		

The pro of puppet stayman is obvious: that it can find 5-3 and 5-4 strains easily. The con is, however, that garbage stayman can no longer be used.

Imagine this: you hold around 0 HCP with singleton club, 4-4-4-1. That's the dream garbage stayman hand, but if you bid 3♣, partner has a high chance of bidding 3NT (and you are screwed, whether you go to 4♦ or not). We can calculate this chance - around 40%. With this in mind, it is no longer beneficial to risk garbage stayman anymore. With 5-4-4-0, if the 5 is in a major, then the chance rises to an even more alarming 47%. Hence, garbage stayman is completely useless with puppet stayman.

4 The Math - 5-3 strains

Let us first see the probability of this choice of whether to use puppet Stayman actually mattering.

In favor of puppet stayman is the probability of a 5-3 strain or a 5-4 strain. There are four conditions: a 5⁺ suit, that suit being major, partner having 5⁺ HCP, and partner having 3⁺ in that suit (minus a 4-3-3-3 with that suit having length 3).

2NT is never opened with unbalanced shape (we'll say that 4-4-4-1 is also unbalanced, although many players like to open 2NT still). That leaves 4-3-3-3, 4-4-3-2, 5-3-3-2, and 5-4-2-2 - I wouldn't open 2NT with 6-3-2-2.

The probability of 5-3-3-2 or 5-4-2-2 is 0.2783321611%, while the probability of any other shape is 0.3659706494%. That is, expect your partner to hold a 5-card suit 43.19896741% of the time, and a 5-card major 10.7997418525% - about 10.80% - of the time. 4-9 HCP occurs 64.29% of the time (for even more HCP, treatments such as 4NT as qualitative invitation can be used instead of puppet stayman). We'll assume the two are independent, since they are pretty close to independent, and we can arrive at 6.9433%. To calculate the probability of a 5-3 strain, we again assume that shape and HCP are independent, and we arrive at 30.5786%.

Before we multiply, consider the probability of holding a 4-3-3-3 - 11.0574%. Then consider the probability that the suit with length 4 is the 5-card major partner has - 2.0477%.

The probability ends up being $(30.5786 - 11.0574 + 2.0477 \cdot 2)\% = 23.6166\%$. Multiplying by 6.9433%, finally, gives 1.640%.

We can therefore analyze the games by generating hands with the following conditions:

1. West has 20-21 HCP
2. West has one of the 4 balanced hand types
3. West has 5⁺ spades
4. East has 4-9 HCP
5. East has 3 spades and not 4-3-3-3, or has 4-3-3-3 with 4 spades

Then, we will analyze the difference between 4♠ and 3NT contracts, with West as declarer. Here are 5 randomly generated examples, in which we will analyze the most logical optimal playing sequence:

Deal 1 Generated Example 1

	♠ 5 4 2	
	♥ 8 5 4 3	
	♦ J 6	
	♣ 9 8 5 2	
♠ A K J 8 6	N W E S	♠ Q 9 3
♥ K T		♥ J 2
♦ A Q 9 8		♦ T 7 4 3
♣ A 6		♣ Q T 7 4
	♠ T 7	
	♥ A Q 9 7 6	
	♦ K 5 2	
	♣ K J 3	

For 3NT, we count: 5 tricks in the easily established spade suit, 1 trick in hearts - but it is very risky to finesse, 3 trick in diamonds due to a successful finesse and 3-2 break, and 1 trick in clubs - ♣K-J is very unluckily right after ♣Q-T.

Here, not calling 3♣ stayman makes north lead hearts (imagining weak majors), easily the best lead given south's ♥A-Q-x-x-x. South takes the trick with ♥A and returns ♥6. Declarer cashes the spade tricks. North gets rid of its useless ♦J-x doubleton, while South makes some safe discards.

Deal 1 ... Continued

	♠ —	
	♥ 8 5	
	♦ —	
	♣ 9 8 5 2	
♠ —	N W E S	♠ —
♥ —		♥ —
♦ A Q 9 8		♦ T 7 4 3
♣ A 6		♣ Q T
	♠ —	
	♥ Q 9 7	
	♦ K 5	
	♣ K J	

South is squeezed here, having to discard another card. Discarding either clubs or diamonds establishes ♣Q or ♦Q, which results in 3NT +3 or 3NT +2. South discards a heart. Declarer plays ♦A, and then gives away a diamond trick to the king. The suit is established, defenders don't have enough hearts to sink the contract, and 3NT is made.

Deal 1 Generated Example 1 again for convenience

	♠ 5 4 2	
	♥ 8 5 4 3	
	♦ J 6	
	♣ 9 8 5 2	
♠ A K J 8 6	N W E S	♠ Q 9 3
♥ K T		♥ J 2
♦ A Q 9 8		♦ T 7 4 3
♣ A 6		♣ Q T 7 4
	♠ T 7	
	♥ A Q 9 7 6	
	♦ K 5 2	
	♣ K J 3	

For 4♠, we count:

1. No losers in spades
2. 1 loser in hearts (♥A)
3. 1 loser in diamonds (♦K or ♦J)
4. 1 loser in clubs (♣K or ♣J)

The rest of the hand is easy to play, 4♠ made.

Notice how 3NT is easily +1 or +2 if the defender makes a slight inaccuracy (e.g. not leading hearts, accidentally (or as a signal) playing ♥8 too early, not returning hearts, etc.), while the declarer can only blunder by not giving away a diamond trick at the end - but there is no reasonable alternative. Meanwhile, 4♠ is very easily defended against.

In an MP match, expect 3NT +1 (430) against 4S = (420), an edge for 3NT. In an IMP match, there is little difference (1 IMP at most). For calculation not done manually, this would be treated as 3NT = against 4S =, since we look at the double dummy evaluation directly. Since it is much easier to error defending a 3NT contract (you don't get second chances to establish your suit), our analysis will be skewed towards puppet stayman here.

Deal 2 Generated Example 2

	♠ K T 7 4	
	♥ 6	
	♦ 9 8 2	
	♣ J 9 8 7 4	
♠ A Q 9 8 2	N W E S	♠ J 6 3
♥ T 4		♥ Q 7 3 2
♦ A J		♦ K T 7 6 5
♣ A K Q 2		♣ T
	♠ 5	
	♥ A K J 9 8 5	
	♦ Q 4 3	
	♣ 6 5 3	

With a freakness as high as 5, south makes the 3NT contract as difficult as he can, especially if partner leads hearts. North, scared that leading from ♠KT74 will lose a trick (it does), leads, miraculously, hearts. If north has exactly two hearts and an outside entry, then declarer is doomed - two finesses, and ♥AK will get his ♥Q. But if north only has one heart, or has at least 3, then we should duck. South plays ♥J - obviously - and leads spades - for partner must have had something in spades to not have led it - hoping to get another heart finesse. ♠5 - ♠2 - ♠7 - ♠J. East plays spade - trying to finesse - but south discards. West, desperate, takes the trick with ♠A. Declarer then bets on taking ♦Q with ♦AK. He cashes ♣AKQ - sees no discards - and ♦A, and then plays ♦J to ♦K. No ♦Q.

Deal 2 ... Continued

	♠ K T	
	♥ —	
	♦ 9	
	♣ J 9	
♠ Q 9 8	N W E S	♠ —
♥ T		♥ Q 7 3
♦ —		♦ T 7
♣ 2		♣ —
	♠ —	
	♥ A K 9 8	
	♦ Q	
	♣ —	

Brilliantly, declarer plays ♦7 from dummy - the only play that can possibly make the contract, regardless of opponent's holdings - and south is thrown in. South is forced to cash in ♥AK and give up ♥8 to dummy's ♥Q. 3NT =.

Deal 2 Generated Example 2

	♠ K T 7 4	
	♥ 6	
	♦ 9 8 2	
	♣ J 9 8 7 4	
♠ A Q 9 8 2	N W E S	♠ J 6 3
♥ T 4		♥ Q 7 3 2
♦ A J		♦ K T 7 6 5
♣ A K Q 2		♣ T
	♠ 5	
	♥ A K J 9 8 5	
	♦ Q 4 3	
	♣ 6 5 3	

For 4♠, the sensible lead of ♥6, followed by ♥AKJ, wins two tricks immediately and threatens to immediately knock out the contract. If west plays ♠x, then north over-ruffs, and otherwise north discards diamonds. Either way, north will win two tricks in trumps, either by ruffing or by taking both ♠K and ♠T. 4♠ -1, and 3NT wins by a large margin.

Deal 3 Generated Example 3

	♠ J 7 6 2	
	♥ K 9	
	♦ T 6	
	♣ Q T 9 5 2	
♠ A K Q 9 3	N W E S	♠ 8 5 4
♥ Q 6 5		♥ 8 7 4 3
♦ A J		♦ 7 4 3 2
♣ K J 6		♣ A 7
	♠ T	
	♥ A J T 2	
	♦ K Q 9 8 5	
	♣ 8 4 3	

3NT -3 while 4♠ -2. 4♠ wins by a large margin. To be honest though, I might pass 3♠ - my hand is too weak, since there's only one finesse possibility and everything else is likely to fall apart as well.

West	North	East	South
2NT	/	3♣	/
3♠	/	/	//

Deal 4 Generated Example 4

	♠ T 5										
	♥ Q 9 8 6 5										
	♦ A K J 8 4										
	♣ 3										
♠ A K Q J 2	<table><tr><td></td><td>N</td><td></td></tr><tr><td>W</td><td></td><td>E</td></tr><tr><td></td><td>S</td><td></td></tr></table>		N		W		E		S		♠ 9 8 6
	N										
W		E									
	S										
♥ A K T 3		♥ J 7 2									
♦ 5 2		♦ T 3									
♣ K 7		♣ A J 9 6 5									
	♠ 7 4 3										
	♥ 4										
	♦ Q 9 7 6										
	♣ Q T 8 4 2										

This is obviously one of the cases in which 3NT goes down terribly, perhaps even doubled. 3NT -1. Declarer has 9 immediate tricks, but cannot stop the diamonds from suffering a disaster. 4♠ = since the tenth trick can be established from ♥T and ♥J by driving the ♥Q out. There are only 3 losers - one in hearts and two in diamonds - so there is no rush.

Deal 5 Generated Example 5

	♠ 8 4 3										
	♥ K 5										
	♦ Q 5 4										
	♣ Q T 5 3 2										
♠ A K 9 7 5	<table><tr><td></td><td>N</td><td></td></tr><tr><td>W</td><td></td><td>E</td></tr><tr><td></td><td>S</td><td></td></tr></table>		N		W		E		S		♠ Q J T
	N										
W		E									
	S										
♥ A Q J		♥ T 6 4 3 2									
♦ 8 7 6		♦ J T 3									
♣ A K		♣ J 7									
	♠ 6 2										
	♥ 9 8 7										
	♦ A K 9 2										
	♣ 9 8 6 4										

3 immediate diamond losers (4 in no trump) exist, and the ♥K cannot be resolved. 3NT -1 and 4♠ -1 as well.

5 The Data - 5-3 strains

By running double dummy on 100,000 boards, we get the following raw result:

		Tricks for 4♠							
		6	7	8	9	10	11	12	13
Tricks for 3NT	1			1	4				
	2		1	6	5	2			
	3			7	25	7			
	4		6	22	39	14	1		
	5	1	36	150	220	83	16		
	6	6	93	814	1567	1084	239	27	
	7	2	107	1568	4997	4840	1538	264	1
	8	2	47	1416	8993	11729	4264	434	1
	9	2	28	408	6171	15126	6034	368	2
	10		9	67	866	8580	7306	688	22
	11			6	86	946	4891	1206	32
	12			1	2	66	348	1561	116
	13					3	36	27	317

Figure 1: Tricks taken for 3NT vs 4♠

It is pretty easy to reach the conclusion that 4♠ has a definitive advantage over 3NT. That's what we expected - we just wanted to quantify it.

There are a few obvious outliers here - let's take a look at them.

Deal 380 Generated Example 380

	♠ 6 5	
	♥ 9	
	♦ 6 5 3 2	
	♣ K Q T 9 5 2	
♠ A K T 4 2	N W E S	♠ J 9 7
♥ A Q 8 6		♥ J T 7 5 4 3
♦ A 8		♦ J 9 7
♣ A 3		♣ J
	♠ Q 8 3	
	♥ K 2	
	♦ K Q T 4	
	♣ 8 7 6 4	

In this hand, we have 7♠ = yet 3NT -2. 19 out of the 25 HCP are immediate winners, but they only supply 5 tricks. Since one has to lose control (♠Q or ♥K) to play from dummy, defender will win 5 club tricks and another trick from elsewhere, contract down 2.

For 7♠, the shortage in clubs of dummy provides not only ruffing potential but also, most importantly, the ability to enter dummy. Even if defender leads diamonds, declarer cashes ♣A, ruffing ♣3 in dummy and finessing one round of spades. If south overtakes with ♠Q, then ♠J is established and declarer draws another round of trumps and leads hearts from the dummy, taking the trick in hand with either ♥Q or ♥A, making the contract by simply cashing sure winners. Otherwise, we continue finessing - this time hearts. It works, and 7♠ is made.

Deal 93791 Generated Example 93791

	♠ 9	
	♥ Q 8 5 4	
	♦ A Q J	
	♣ Q 7 4 3 2	
♠ A K J 6 5	N W E S	♠ T 7 3
♥ A 3 2		♥ K J T 9 7 6
♦ K 4 2		♦ 8 7
♣ A J		♣ T 8
	♠ Q 8 4 2	
	♥ —	
	♦ T 9 6 5 3	
	♣ K 9 6 5	

In this hand, we have quite the opposite: 4♠ -2 yet 6NT can be made. Here, south's void hearts makes ruffing easy. Additionally, defenders can

finesse diamonds. In the end, north can win 2 diamond tricks before he gets ruffed, and south will manage to ruff 3 times - once for the lead, and once more for each diamond entry north gets. For 6NT, not only is no ruffing available, but south also lacks any entry ($\clubsuit K$ being closely guarded by $\clubsuit A$). Declarer can establish 5 tricks from spades and 6 tricks from hearts, along with $\clubsuit A$, 6NT =. All the defenders can do is take $\diamond A$ and accept declarer's claim after his finesses work.

Overall, 55.321% of the hands were made in 3NT, whereas an overwhelming 72.219% of the hands were made in $4\spadesuit$. In fact, only 8.71870 tricks can be taken in NT on average, while for spades it goes up to 10.02356 - still not promising, but well above the threshold. In fact, only 2.952% of the hands yield more tricks in no trump, 23.049% are the same, and spade contracts are superior 73.999% of the time. Even if we give one less trick to spade contracts - after all, they need one extra trick to make game - the stat is at 26.001% : 34.409% : 39.590%. The MP score (regardless of vulnerability - the reader can verify this by themselves) come at 26.001% : 10.655% : 63.344%. By points, $4\spadesuit$ wins by 98.1568 (non-vulnerable) and 149.3908 (vulnerable) on average - most of this is from under-tricks. In terms of IMPs, $4\spadesuit$ wins by 2.32271 and 2.75913, respectively - not as big of a difference as it was thought to be. We'll be looking at IMP, the most common form of scoring in competitive bridge, in this analysis, but other data will be provided for posterity. Multiply 1.640% (the chance the choice of whether to use puppet stayman actually matters) with 2.32271 and 2.75913, respectively (the expected gain), we get 0.038 and 0.045, respectively.

6 The Math - Garbage Stayman

Garbage stayman can be bid with 0 to 3 HCP with a relative shortage in clubs and no 5-card major. 0 to 3 HCP has a chance of 16.64% to occur.

Our criteria for the shape is: 3 or 4 cards in each major (more than 5 would require $3\Diamond$ or $3\Heart$ transfer bids) and 4 or 5 cards in diamonds (since $3\Diamond$ response to $3\clubsuit$ doesn't promise length in diamonds). That has probability 9.45%. Multiplying yields 1.57248% - pretty similar compared to the 1.640% in the last section.

Let us look at a set of such hands:

Deal 1 Generated Example 1

♠ A Q 6 2	♠ 9 5 4	♠ J T 7
♥ A J 9	♥ 5 3 2	♥ T 8 6 4
♦ K Q 8 6	♦ A 3	♦ T 9 7 4
♣ A 9	♣ Q J 4 3 2	♣ 7 6

♠ K 8 3

♥ K Q 7

♦ J 5 2

♣ K T 8 5

Here north leads clubs against 2NT - the obvious choice - and declarer ducks. Club is returned and taken by declarer's ♣A. Declarer has to establish diamond winners fast, and leads ♦K. Unfortunately for declarer, defenders play ♣4 correctly (♣J might confuse partner into not overtaking with ♣K, costing a trick) and all the clubs tricks are taken. Declarer has to concede a trick to one of ♥KQ, 2NT -1.

Meanwhile, $3\spadesuit$ is quite safe with only one trick lost in clubs instead of four. No loser in spades, one loser in hearts, one loser in diamonds, and one loser in clubs. In practice, though, there is a loser in spades since dummy doesn't have enough entries to finesse enough times in spades. $3\spadesuit =$.

Deal 2 Generated Example 2

	♠ 9 7 6 3										
	♥ J 6 5										
	♦ K Q 8 7										
	♣ T 5										
♠ K T	<table border="1"><tr><td></td><td>N</td><td></td></tr><tr><td>W</td><td></td><td>E</td></tr><tr><td></td><td>S</td><td></td></tr></table>		N		W		E		S		♠ Q 5 4
	N										
W		E									
	S										
♥ A K 7		♥ T 8 3 2									
♦ A 6 4 2		♦ J 9 5 3									
♣ A Q J 3		♣ 7 6									
	♠ A J 8 2										
	♥ Q 9 4										
	♦ T										
	♣ K 9 8 4 2										

In 2NT, defender leads diamonds. Declarer wins with ♦J in dummy and plays ♣6 to finesse - it works. Declarer then plays ♠K from hand (trying to establish ♠Q and get an entry). South ducks, and declarer fakes finessing ♠Q, which doesn't work. South cashes his two spade winners, with declarer trashing a diamond and a club. North plays ♠9 before ♠7 - he doesn't want to finesse against his partner. South returns hearts and declarer cashes ♥AK and leads a small heart to establish the suit - south is thrown in. Declarer gets the ♦A and ♣A-Q tricks, 2NT =.

With 3♦, there are 4 losers: ♠A, ♥J, and ♦K-Q. 3♦ =. 2NT technically wins in MP but the deal is a tie in IMP.

Deal 3 Generated Example 3

	♠ Q 8 6										
	♥ J T 7 6 2										
	♦ K 3										
	♣ T 6 4										
♠ J 9	<table><tr><td></td><td>N</td><td></td></tr><tr><td>W</td><td></td><td>E</td></tr><tr><td></td><td>S</td><td></td></tr></table>		N		W		E		S		♠ T 7 4 2
	N										
W		E									
	S										
♥ A K Q 4		♥ 8 5 3									
♦ A J 4		♦ Q 9 8 2									
♣ A J 8 5		♣ 7 2									
	♠ A K 5 3										
	♥ 9										
	♦ T 7 6 5										
	♣ K Q 9 3										

2NT -1 due to lack of entries and possible winners, while 3♥ -3 due to all of the above and a 5-1 split in trumps.

Deal 4 Generated Example 4

	♠ Q J	
	♥ A Q 3 2	
	♦ K 3 2	
	♣ T 8 6 2	
♠ A K 6 5 2	N W E S	♠ 9 7 3
♥ K 7 6		♥ J T 9 8
♦ A 8		♦ J T 9 7 5
♣ A Q 7		♣ 5
	♠ T 8 4	
	♥ 5 4	
	♦ Q 6 4	
	♣ K J 9 4 3	

One of the best hands to call garbage stayman. One loser in spades (♠Q-J-T), two losers in hearts, one loser in diamonds and a scary club formation makes 2NT hard to play. In fact, declarer can't prevent the 3 eventual loser in clubs but can only try and eliminate other losers before that. Optimally, defenders has time to take 6 tricks and 2NT -1.

In a 3♠ contract, there is the same one loser in spades, two losers in hearts, and one loser in diamonds, but the clubs losers are eliminated by dummy's singleton. 3♠ = easily.

Deal 5 Generated Example 5

	♠ A J 9 5	
	♥ J 5	
	♦ 6 5 2	
	♣ K 9 5 2	
♠ K 8	N W E S	♠ T 6 3 2
♥ A Q 9 7 4		♥ T 8 3
♦ A K T 8		♦ Q J 9 7 3
♣ A 6		♣ 8
	♠ Q 7 4	
	♥ K 6 2	
	♦ 4	
	♣ Q J T 7 4 3	

Here, 2NT just barely makes - declarer has just enough immediate winners (♥A-Q, ♦A-K-Q-J-T, and ♣A) to make the contract before clubs become a complete disaster. 3♥, on the other hand, has +1, since one spade and one heart is all declarer has to lose.

7 The Data - Garbage Stayman

By running double dummy on 100,000 boards, we get the following raw result:

		Tricks for 3♦/3♥/3♠										
		3	4	5	6	7	8	9	10	11	12	13
Tricks for 2NT	0					1						
	1						4	1	1			
	2			1	3	10	15	15	3			
	3	2		10	33	66	73	55	8	1		
	4	3	30	89	292	433	303	148	39	3		
	5		35	480	1723	2466	2077	919	167	17	1	
	6	1	44	384	3012	6743	6528	3595	863	154	9	
	7	1	18	154	1429	7741	11939	8517	2717	393	37	4
	8	1	5	50	401	2309	8691	9834	4566	864	51	
	9			11	45	289	1146	3360	2576	616	26	
	10				5	15	57	227	553	326	35	5
	11						1	10	19	55	29	6
	12									1	3	2

Figure 2: Tricks taken for 2NT vs 3♦/3♥/3♠

There is obviously a lot more variation. We'll take a look at three outliers first.

Deal 93266 Generated Example 93266

	♠ 9 5 4										
	♥ A K J T 4 2										
	♦ T 3										
	♣ A T										
♠ Q J	<table border="1"><tr><td></td><td>N</td><td></td></tr><tr><td>W</td><td></td><td>E</td></tr><tr><td></td><td>S</td><td></td></tr></table>		N		W		E		S		♠ 8 6 2
	N										
W		E									
	S										
♥ Q 7		♥ 9 6 3									
♦ A K Q J 7		♦ 9 8 5 4									
♣ K Q 5 3		♣ 9 6 4									
	♠ A K T 7 3										
	♥ 8 5										
	♦ 6 2										
	♣ J 8 7 2										

In this deal, no tricks can be taken by the declarer in NT while 7 can be taken if garbage stayman is called (and 3♦ is reached).

It is pretty clear why an NT contract is doomed: ♥A-K gets the ♥Q and ♠A-K gets ♠Q-J, and the majors provide 11 tricks to the defender. Coupled with ♣A and some endgame plays, the defenders take all 13 tricks. 2NT -8, although the practical sequence is probably for north to double (since he can lead hearts and maintain a club entry regardless of whether he can get the ♥Q) and for east to either call diamonds directly or redouble as SOS in some systems.

In 3♦, there is at least one club trick and 5 diamond tricks to be taken by declarer. There's also potential for ruffing, which provides an extra tricks. 3♦ -2.

Deal 56594 Generated Example 56594

	♠ Q J 2										
	♥ 9 6 2										
	♦ 8 5 3										
	♣ K 4 3 2										
♠ K 7	<table border="1"><tr><td></td><td>N</td><td></td></tr><tr><td>W</td><td></td><td>E</td></tr><tr><td></td><td>S</td><td></td></tr></table>		N		W		E		S		♠ 9 6 5 3
	N										
W		E									
	S										
♥ A K Q J 7		♥ 8 5 4 3									
♦ A Q 7		♦ J T 9 6 2									
♣ J 9 5		♣ —									
	♠ A T 8 4										
	♥ T										
	♦ K 4										
	♣ A Q T 8 7 6										

Here's a similar deal. It is also rather clear why NT fails - six glaring club winners to the defenders, along with ♠A. There is also a spade loser for declarer, since dummy has no entry to finesse. 2NT -3.

The play for the makeable 6♠ is simple, since now east's void clubs allow him to ruff and declarer can now play spades from dummy.

Deal 33968 Generated Example 33968

	♠ 4										
	♥ J 8 7 6 5 4										
	♦ A J										
	♣ K 7 6 5										
♠ A J T 5	<table><tr><td></td><td>N</td><td></td></tr><tr><td>W</td><td></td><td>E</td></tr><tr><td></td><td>S</td><td></td></tr></table>		N		W		E		S		♠ 9 8 2
	N										
W		E									
	S										
♥ A T 9		♥ Q 3 2									
♦ K Q 2		♦ T 9 6 5 3									
♣ A Q 9		♣ T 4									
	♠ K Q 7 6 3										
	♥ K										
	♦ 8 7 4										
	♣ J 8 3 2										

In this deal, the no-trump contract has the advantage. The unfortunate 4-3 strain of spades and 5-1 split is enough misfortune, and south's singleton hearts eliminates ♥Q as a possible entry to dummy. 3♠-1.

2NT can obtain 3 overtricks since now ♥Q is indeed an entry to dummy and a chance to finesse. Additionally, diamonds can now be established. Declarer counts three tricks in spades (finessing twice from dummy), three tricks in hearts (♥A drops ♥K), four tricks in diamonds (losing one to ♦A), and one trick in clubs - 11 tricks. 2NT +3.

Overall, 36.190% of the hands were made in 2NT, whereas a much better 40.831% of the hands were made in $3\Diamond/3\heartsuit/3\spadesuit$. 32.581%, 37.889%, and 29.530% of the hands played $3\Diamond$, $3\heartsuit$, and $3\spadesuit$, respectively, with a make rate of 43.983%, 38.864%, and 39.878%. Only 7.04696 tricks can be taken in NT on average, while for trump contracts it goes up to 8.19780 (individually 8.255, 8.155, and 8.189). The MP score come at 35.241% : 20.494% : 44.265% (slightly in favor of garbage stayman, of course). Garbage stayman has an edge of 16.6838 points non-vulnerable or 23.4893 points vulnerable - a small margin - which translates to 0.54513 IMPs or 0.63699 IMPs, respectively. Multiplying by 1.57248% gives the final yield of 0.009 or 0.010, respectively. Compare that with 0.038 and 0.045 in the last study.

Puppet Stayman, therefore, is definitely the better choice.

8 Conclusion

Puppet Stayman has proved itself to yield around an expected 0.03 IMPs per game of 2NT opening, which occurs with probability 0.657%. That's an average of about 0.0002 IMPs per game.

To put this into perspective, 100 hands a day is a remarkable statistics (kept long enough). You'll expect to gain 6 IMP every year playing Puppet Stayman. An average player plays around 100 hands per week at most. Expect to gain 1 IMP every year.

Puppet Stayman, however, is not the easiest of conventions. Compared to Roman Key Card Bids, Texas Transfers, or 2-over-1 Game Forces, it is much more nuanced in its continuations. Players, especially ones at entry level or even amateur level, do not need to worry about Puppet Stayman, since a single mistake can cost much, much more than 0.03 IMP.

If opener has 5-4 majors (specifically 5 spades and 4 hearts), then we encounter another problem. System designers have designs systems in which $3\heartsuit$ and 3NT responses to 2NT - $3\clubsuit$ are switched, but this convention is easy to forget, which can cause massive problems - again, problems much more serious than 0.03 IMP.

Additionally, even for experienced players, one major caveat of the Puppet Stayman system is that it gives away a lot more information than normal Stayman, and playing bridge is more than looking up a double-dummy table, and this is a major problem especially if responder calls Puppet Stayman with any hand with a 3-card major, e.g. $\spadesuit J \times \times \heartsuit \times \times \diamondsuit A \times \times \times \clubsuit \times \times \times$, where it is obviously (or non-obviously) absurd to do so (there is no downside to use Puppet Stayman in our study since we use double dummy analysis, so Puppet Stayman is always used in these scenarios). This poses another risk in using Puppet Stayman.

Many resources online argue in favor of Puppet Stayman, but most of these resources are not high in quality, and the majority of the remaining ones don't provide any counterarguments, only pointing out the positive effects of Puppet Stayman.

It is debatable how much better Puppet Stayman will perform in real-life scenarios (if at all). For now, though, it is recommended to put off Puppet Stayman if it is not something you have already familiarized yourself with - 0.03 IMP is not substantial.